AgriHelp – Your Smart Farming Companion

System Design Project Proposal – by Avik Srivastava(24BCS10643)

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1. Introduction

Agriculture today faces challenges such as unpredictable weather, crop diseases, lack of expert guidance, and limited market transparency. AgriHelp is an AI-powered agricultural assistance platform designed to support farmers by providing crop recommendations, disease detection, market price insights, and community knowledge sharing through a simple and accessible digital system.

The platform supports multiple Indian languages, and integrates AI capabilities using the Gemini API to deliver intelligent, context-aware assistance.

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2. Technology Stack

- **Frontend:** React (Mobile-first, PWA)
- **Backend:** Spring Boot (REST APIs)
- **Database:** MongoDB
- **AI Services:** Gemini API
- **External APIs:** Weather data, mandi price feeds

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3. High-Level System Architecture

At a high level, AgriHelp follows a simple layered architecture:

User

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React Frontend

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Spring Boot Backend

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MongoDB

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Gemini API + External APIs

- The frontend handles user interaction and input.
- The backend processes requests, manages security, and coordinates AI services.

- MongoDB stores all application data.
- Gemini API provides AI intelligence for all smart features.

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4. Module Breakdown (8 Modules)

🔹 **Frontend Modules (1–4)**

Module 1: User Interface & Navigation

Provides a simple, responsive, and mobile-friendly interface for farmers.

- Dashboard-based navigation
- Offline support using PWA
- Easy access to AI assistant and community features

Module 2: Farmer Profile & Language Selection

Manages farmer details and personalization.

- Location, land size, soil type
- Preferred language selection
- Data used to personalize AI responses

Module 3: AI Assistant Interface

Acts as the main interaction point with AI.

- Chat-based interface
- Voice input/output
- Image upload for plant disease detection

Module 4: Community & Discussion Forum

Enables farmers to learn from each other.

- Create posts and comments
- Crop-specific discussions
- Knowledge sharing and peer support

🔹 Backend Modules (5–8)

Module 5: Authentication & User Management

Handles secure access to the platform.

- User registration and login
- JWT-based authentication
- Profile management APIs

Module 6: AI Intelligence Service

Core intelligence layer powered by Gemini API.

- Crop recommendations
- Disease identification
- Multilingual chatbot responses
- Market reasoning support

Module 7: Market Price & Forecast Service

Provides market-related insights.

- Current mandi prices
- Short-term (7-day) price forecasts
- Location-based results

Module 8: Database & Data Management

Stores and manages all system data.

- User profiles
- AI interactions
- Market data
- Community posts

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5. Low Level Design (LLD – Brief)

The Low Level Design defines how individual components interact internally.

- **Frontend LLD:**
  React follows a component-based structure with separate pages for Dashboard, Assistant, Community, and Profile. API calls are handled through service files, and global state (authentication and language) is managed using Context API.
- **Backend LLD:**
  Spring Boot follows a controller-service-repository pattern. Each feature has dedicated REST endpoints. Authentication is handled using JWT filters, and AI requests are routed through a centralized AI service that communicates with the Gemini API.
- **Database LLD:**
  MongoDB collections are used for users, AI queries, market prices, and community posts, enabling flexible and scalable data storage.

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6. Non-Functional Requirements

- **Scalability:** Supports a large number of users
- **Security:** Encrypted authentication tokens
- **Usability:** Simple UI for non-technical users
- **Accessibility:** Multilingual and voice support
- **Reliability:** Works in low-network conditions

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7. Conclusion

AgriHelp is a **simple, scalable, and AI-driven agricultural assistance platform** designed to make expert knowledge accessible to farmers. By combining modern web technologies with AI through the Gemini API, the system provides practical decision support while remaining easy to understand, implement, and maintain.